MIL-C-25200B (USAF)
1 October 1971
Superseding
MIL-C-25200A (USAF)
9 April 1962

MILITARY SPECIFICATION

CABLE ASSEMBLIES, SPECIAL WEAPONS, ELECTRICAL, GENERAL REQUIREMENTS FOR

1. SCOPE. This specification establishes the detailed requirements for the design, manufacture and testing of Air Force special weapons cable assemblies, including interconnecting cables. (See 6.1)

2. APPLICABLE DOCUMENTS. The following documents, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

HH-I-595	Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic, for Low-Temperature Application
QQ-S-571	Solder; Soft (Tin, Tin-Lead, and Lead-Silver)

Military

MIL-W-76	Wire and Cable, Hook-Up, Electrical Insulated
MIL-V-173	Varnish, Moisture- And Fungus-Resistant, For the Treatment of Communications, Electronic and Associated Electrical Equipment
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-T-713	Twine, Lacing and Tying, Electrical and Electronic Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-I-7444	Insulation Sleeving, Electrical, Flexible
MIL-S-8516	Sealing Compound, Synthetic Rubber, Accelerated (Electric Connectors and Electric Systems)
MIL-F-14256	Flux, Soldering, Liquid (Resin Base)

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MIL-W-16878	Wire, Electrical (Insulated, High Temperature)
MIL-D-26715	Descriptive Identification Data to be Furnished by Government Suppliers
MIL-C-27599	Connector, Electrical, Miniature, Quick Disconnect (For Weapons Systems) Established Reliability
MIL-C-38999	Connectors, Electrical, Quick Disconnect, Removable Crimp Type Contact, Reliability Assurance Program

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods for Aerospace Ground Equipment
MS 3420	Bushing-Cable, Adapter, Telescoping
MS 27342	Adapter, Electrical Connector
MS 27495	Tool, Contact, Connector, Assembly and Disassembly
MS 27509	Tool, Contact, Connector, Assembly and Disassembly

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PRECEDENCE OF STANDARD MATERIALS, PARTS AND PROCESSES. Any material, standard part or process not covered herein shall comply with specifications and standards selected in the order of precedence set forth in MIL-STD-143.

3.1.1 COMMERCIAL PARTS. Commercial parts equal or superior to standard (MS or AN) items may be used when, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

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b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.1.2 STANDARD PARTS. With the exception in 3.1.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.2 SUBSTITUTION. If a contractor desires to substitute another part, material, process or design specified by a previously approved specification or drawing, he shall submit to the procuring activity a statement describing the proposed substitution and his reasons for desiring to substitute. He shall also submit evidence that the substitution is equal or superior to the specified item. The procuring activity may require samples for the determination or suitability. Substitutions shall not be made without the prior written approval of the procuring activity.

3.3 MATERIAL

3.3.1 FLAMMABLE MATERIALS. Flammable materials which will ignite or explode from an electrical spark or from heating shall not be used.

3.3.2 TOXIC AND CORROSIVE FUMES. As installed, and under the environmental conditions specified herein, the materials shall not liberate gases which combine with the atmosphere to form acids or corrosive environments nor shall they liberate toxic fumes when burned.

3.3.3 FUNGUS-INERT MATERIALS. Materials which are non-nutrients for fungi shall be used to the greatest extent practicable.

3.3.3.1 FUNGUS-NUTRIENT MATERIALS. Unless otherwise specified, when a fungus-nutrient material is specified, or is the only satisfactory material that may be used, it shall be treated with a coating of varnish in accordance with MIL-V-173 (with fungicide).

3.3.4 CORROSION-RESISTANT MATERIALS. All materials used in the assembly shall be corrosion resistant or treated to resist corrosion.

3.3.4.1 ADDITIONAL PROTECTION OF BASE METAL. All copper conductors shall be tinned, and all connectors and wire terminations shall be suitability coated, tinned, or solder-dipped.

3.3.5 LUBRICANTS. Lubricants shall be used sparingly. When a lubricant is necessary to facilitate assembly of the covering over a group of wires, it is recommended that pure powdered mica or talc be used. To resolve problems of excessive adhesion which may occur during assembly of the back grommet in C or M type, or equal, MS connectors, or during the covering of a long cable composed of a large number of wires, the use of nonhygroscopic silicone grease is permitted. When used, silicone grease should be applied as a thin film by means of a grease-impregnated cloth.

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3.4 DESIGN. The contractor shall design the cable assembly, subject to the requirements of this specification, in such a manner as to facilitate quantity production.

3.4.1 WIRES AND CABLES. Unless otherwise specified, the component wires and cables shall be type MW, HW, or equal with construction in accordance with the requirements of MIL-W-76 or MIL-W-16878 as applicable. Each conductor shall be stranded, and the equivalent size shall be not smaller than number 20 AWG.

3.4.2 SLEEVING. The sleeving shall be clear polyvinyl chloride tubing in accordance with MIL-I-7444.

3.4.3 LACING CORD. Nylon cord in accordance with MIL-T-713, type P, Class 2, is preferred. The size and weight of cord used shall be appropriate to the cable size at the point of application. Cord must be laced in such a manner that if a break in the cord develops, the entire lacing will not be affected.

3.4.4 TAPE. Unless otherwise specified, the cable shall be taped with wrapping tape of polyvinyl chloride in accordance with HH-I-595, or protected with tubing conforming to MIL-I-631, Type F, Form U, Grade C, Class I, Category I.

3.4.5 LIQUID CEMENT. The cement shall be 20 percent by weight of polyvinyl chloride in cyclohexanone, or tetrahydorfuran and a vinyl resin. The latter composition shall be 20 percent by weight of resin (15g/resin and 85 gm/solvent).

3.4.6 SOLDER AND FLUX. The solder shall be SN-60 or SN-50 in accordance with QQ-S-571, either plain or fluxcore. The flux shall be resin in conformance to Specification MIL-F-14256.

3.4.7 CONNECTORS. Unless otherwise specified, the connectors shall be MS type or equal in accordance with MIL-C-27599 (USAF) or MIL-C-38999 (USAF).

3.4.7.1 ADAPTERS. Unless otherwise specified, appropriate cable adapters shall be in accordance with MS27342.

3.4.7.2 BUSHINGS. Unless otherwise specified, cable clamp bushings shall be in accordance with MS3420.

3.4.7.3 DUST CAPS. Dust caps shall be inserted in, or placed over, all MS connectors to prevent contamination of insulators and contacts during transit and storage. Polyethylene caps are preferred.

3.4.8 COVERING. All compound (multiwire) cables shall be covered with sleeving from connector to connector, connector to branch, and branch to branch.

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3.4.8.1 CABLE ENDS. For MS connectors, the external covering shall be extended through the bushing and cable clamp and shall be held in place only by the clamp and bushing. When clips or terminal lugs are used, the sleeving shall be secured by binding in accordance with the requirements of this specification for lacing and taping. An approved alternate method for connectors without bushings permits the sleeving to extend beyond the cable clamp washer, in which case the end shall be secured in accordance with the requirements of this specification for lacing. An approved alternate method for use with MS connectors shall be potting.

3.4.8.2 JUNCTIONS. The junction of main and branch cables shall be laced (step 1 of figure 1) to secure the ends of the branch sleeves. The main cable covering shall then extend 1 1/2 inches over the ends of the branch sleeving (step 2 of figure 1), and shall be laced and taped in accordance with the requirements of this specification for lacing and taping (steps 2 and 3 of figure 1). The approved alternate method (steps A through D of Figure 1) permits the branch sleeves to be butted against the main cable sleeve. This junction shall then be laced as in step B, after which a 3-inch polyvinyl sleeve shall be centered over the joint and secured at both ends in accordance with the requirements of this specification for lacing and taping and steps C, and D of figure 1, or protected with a tubing conforming to MIL-I-631, Type F, Form U, Grade C, Class I, Category I.

3.4.8.3 INTERCHANGEABILITY. Cables performing identical functions shall be completely interchangeable.

3.4.9 MS CONNECTOR ATTACHMENT

3.4.9.1 PREPARATION. Preparations for attaching an MS connector to a cable assembly shall include:

a. Inserting the cable end through the cable clamp adapter and bushing.

b. Stripping and tinning the wire ends. (Mechanical stripping methods shall not be used).

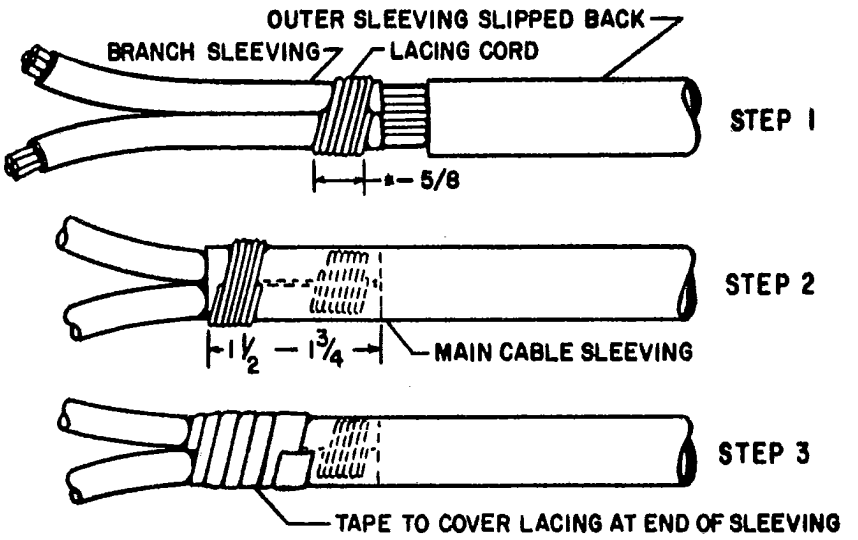
c. Inserting the wire ends through the back grommet for C and M type, or equal, connectors, or through one-inch lengths of vinyl tubing in accordance with 3.4.2 for other types.

3.4.9.2 ATTACHMENT

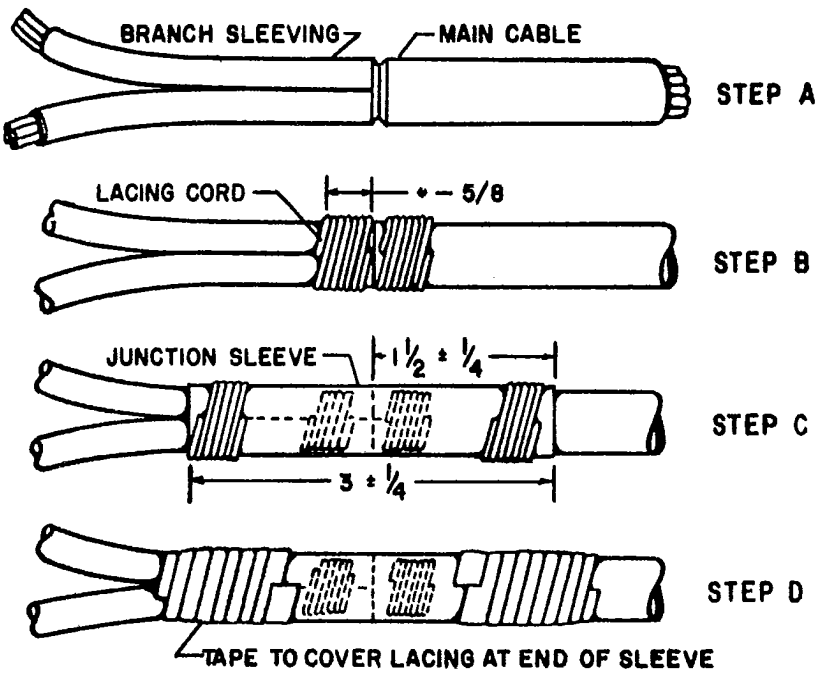
3.4.9.2.1 SOLDER TYPE. Following the preparatory operations each wire shall be securely soldered into its proper solder cup according to the schematic or wiring diagrams. After the soldering has been completed, the back grommet for C and M types shall be slipped over the

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PREFERRED METHOD



ALTERNATE METHOD



NOTE: 1. SECURE ENDS OF LACING CORD WITHOUT KNOTS

2. DIMENSIONS IN INCHES.

* SEE PARAGRAPH 3.9.3

FIGURE 1 CABLE JUNCTIONS

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solder cups and cemented to the cups with vinyl base cement. For R type, a polyvinyl sleeve shall be placed over the connector contacts. The cable clamp and bushing shall be positioned and the compression member drawn down firmly so that the tension applied to the cable shall be transmitted to the connector shell through the cable clamp adapter and not through the terminals.

3.4.9.2.2 CRIMP TYPE. Individual contacts will be inserted and extracted with the MS 27495 or MS 27509 tools. The engaging end of the pin contacts in assembled connectors shall be located within 0.024-inch diameter of true position and the engaging end of socket contacts in assembled connectors shall be located within 0.015-inch diameter of true position. The contact locking device is contained in the insert and will retain the contact so as to meet the retention requirements of MIL-C-38999.

3.4.10 POTTING

3.4.10.1 PREPARATION

3.4.10.1.1 CONNECTOR PREPARATION. Care shall be exercised to maintain the cleanliness of the connectors. Unless otherwise specified on the product drawing, the internal surfaces and/or insert shall not be cleaned or otherwise treated with solvents. Cleaned surfaces shall not be handled with bare hands. They shall be kept free from contamination and shall be handled in a manner to prevent their becoming contaminated. All solvents used for cleaning shall be reagent grade or equivalent.

3.4.10.1.1.1 CONNECTOR SHELL. Thread area of the connector shell shall be cleaned by wiping with a clean lint-free cloth or swab moistened with trichloroethylene, methyl ethyl ketone (MEK) or toluene. Care shall be exercised to prevent entrance of solvent into the internal portions of the connector.

3.4.10.1.1.2 ADAPTER RING. The adapter ring shall be removed and solvent cleaned with trichloroethylene by one of the following methods:

a. Vapor degreasing

b. Ultrasonic cleaning

c. Wiping with a clean cloth or paper wipe moistened with clean trichloroethylene.

Unless otherwise specified on the cable drawings, the adapter ring shall be installed on the connector shell fingertight. If tightening the adapter ring prevents free rotation of the connector coupling ring, the requirements of Figure 2 shall govern. The adapter ring may be bonded in place using the base potting compound. When specified on the product drawing, the adapter ring shall be sealed and bonded to the connector shell with a bead of sealing compound per MIL-S-8516. The sealing compound shall be permitted to gel before further processing.

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3.4.10.1.1.3 When connectors with plastic inserts are used (example MS type) after soldering the conductors, the rear face of the insert shall be sealed with a thin coating of polysulfide per MIL-S-8516 in order to prevent leakage of potting compound into internal parts.

3.4.10.1.2 CABLE PREPARATION. The cable wiring shall be formed to produce either a straight or angle assembly in accordance with the product drawing (see figure 2).

3.4.10.1.2.1 LACING. Unless otherwise specified in the cable drawing, the cable shall be laced as shown in Figure 2 with lacing material per MIL-T-713, Grade P, Class 2. The lacing shall not distort or displace the connector contacts.

3.4.10.1.2.2 CONNECTOR ORIENTATION. Unless otherwise specified on the cable drawing, orientation of the connector key or keyway is optional.

3.4.10.1.2.3 CENTERING OF CABLE. Unless otherwise specified on the cable drawing, the wire bundle or cable shall be centered within the mold in accordance with the requirements of Figure 3 and the following paragraphs. It is recommended that the wall thickness of encapsulant be at least 0.12 inches.

a. If dimension "A" minus dimension "B" is 0.18 inch or less, then dimension "C" shall not vary by more than 0.03 inch.

b. If dimension "A" minus dimension "B" is greater than 0.18 inch, but not greater than 0.31 inch, then dimension "C" shall not vary by more than 0.06 inch.

c. If dimension "A" minus dimension "B" is greater than 0.31 inch, then dimension "C" shall be a minimum of 0.12 inch.

3.4.10.1.2.4 SLEEVING. Unless specified on the cable drawing, insulation sleeving shall not be used over the connector contacts.

3.4.10.1.3 POTTING COMPOUND. The potting compound shall consist of materials equivalent to or better than those listed below:

a. Base Compound (silicon filled epoxy resin).

(1) Hysol Corporation base compound, or;

(2) Furane Plastics Corporation base compound.

b. Curing Agent (Amine)

(1) Hysol Corporation Hardener No. 3471

(2) Furane Plastics Corporation No. HN 9410.

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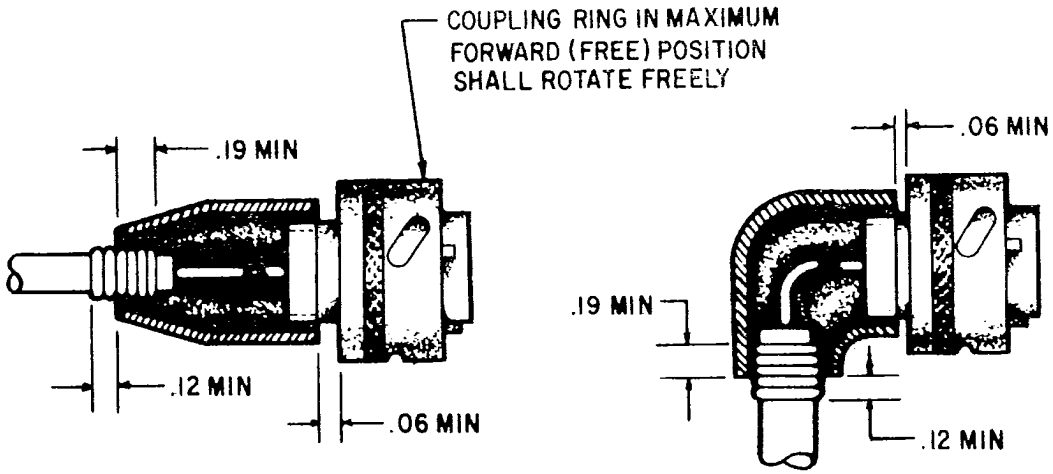


FIGURE 2 PREPARATION OF CABLE ENDS FOR POTTING

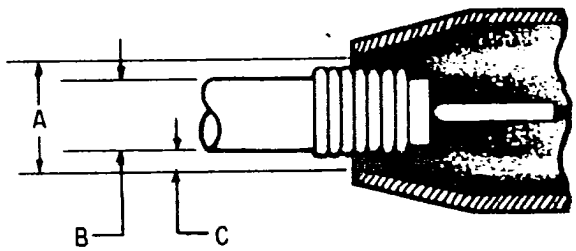


FIGURE 3 CENTERING OF CABLE END

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3.4.10.1.3.1 FORMULATION. Tolerance on weights of materials shall be plus or minus 1 percent.

Base Compound	100 parts by weight
Curing Agent	14.5 parts by weight

NOTE: The work life of this potting compound at 75°F is approximately 35 minutes from the time the curing agent is added.

3.4.10.1.3.2 MIXING. The filled base-resin shall be thoroughly mixed in its shipping container until all settled filler has been dispersed and the mixture is uniform throughout. The required amounts of each component of the compound shall be mixed as outlined below. It is recommended that the material be prepared in batches no larger than one quart to obtain maximum pot life. Use a container whose diameter is approximately equal to its height and whose volume is at least four times the volume of the mixed compound.

3.4.10.1.3.3 Add the hardener to the filled base-resin mixture. Mix thoroughly for several minutes until the mixture is uniform. Material shall be scraped from the sides of the container and from the mixing blade periodically during mixing, and incorporated in the center of the mass. If a mechanical mixer is used, care shall be taken to minimize the amount of heat generated within the mix since heat will reduce the pot life of the compound.

CAUTION: To prevent undue exposure to vapors there should be adequate ventilation during the handling of the resins and hardeners of these potting materials. Ingestion or skin contact with these materials should be avoided. Should accidental skin contact occur, the exposed areas should be washed immediately with mild soap and water.

3.4.10.1.3.4 DEAERATION. As soon as mixing is complete, evacuate the compound at a pressure of 0.5 to 3.0 mm hg absolute until the foam rise collapses. If the mixture is transferred into a pressure injection container, this procedure shall be repeated.

3.4.10.1.4 MOLD PREPARATION. Molds shall be clean, dry, and free from wax, oil, and grease. When the mold is not an integral part of the assembly, it shall be treated with a suitable mold release agent.

3.4.10.2 POTTING.

3.4.10.2.1 PREPOTTING. Unless otherwise specified on the product drawing, after the conductors are soldered in place, but prior to lacing, fill the back end of the connector approximately level with the top of the connector shell with potting material. Mechanically work the potting compound to minimize the presence of voids. Allow the compound to cure before further processing. Cure times shall be a minimum of four hours at room temperature.

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3.4.10.2.2 POURING. The connector mold shall be filled with potting compound in a manner that will minimize air entrapment. Molds may be capped with additional resin at any time before completion of the room temperature portion of any of the allowable cure cycles. Prior to capping, the exposed resin surface shall be kept free from contamination. After capping, molds shall be given a full cure in accordance with the cure schedules of this specification. Care should be exercised in handling loose jacketed cable bundles to prevent sucking up potting compound.

3.4.10.2.2.1 Any excess potting compound on the outside of the connector or mold shall be removed with acetone or methyl ethyl ketone before it has hardened.

3.4.10.2.2.2 Bubbles appearing on the surface of the potting compound prior to gelation shall be broken by spraying with resin defoamer equivalent to or better than Par Industries Inc. Resin Defoamer No. 1, or with isopropyl or ethyl alcohol using an atomizer type sprayer.

3.4.10.3 CURING. The potting compound shall be cured in accordance with one of the cure schedules shown below. The detail specification shall specify cure schedule to be used. All schedule times are minimum times.

CURE SCHEDULES

	1 (NOTE 1)		2 (NOTE 2)		3 (NOTE 3)	
	Time (Hours)	Temp* (°F)	Time (Hours)	Temp* (°F)	Time (Hours)	Temp* (°F)
Initial Cure	4	RT**	2	RT	2	RT
Post Cure	44	RT	2	120±5	1	160±5

* Mold line temperature
** Minimum room temperature 65°F

NOTES: 1. The initial cure of schedule 1 is the minimum time allowed before the part may be removed from the mold. With small connectors having a large amount of heat sink material present, the initial cure should be increased up to 8 hours. The total cure time, however, shall remain 48 hours.

2. For schedule 2 or 3, the part shall not be handled or removed from the mold before completion of the post cure.

3.4.10.4 FINISHED TERMINATIONS. Wire or connections shall not be visible through the potting material. Surface bubbles, bubbles at mold interface, or voids shall not be over 0.09 inch in depth. There shall be no visible cracks or fissures in the cured potting compound.

3.4.11 JUMPERS. When jumpers are specified they shall be of sufficient length to clear all other terminals or cable clamps and shall be of insulated wire conforming to MIL-W-76 or MIL-W-16878

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3.5 PERFORMANCE

3.5.1 CONTINUITY. For the finished cable assembly, electrical continuity shall exist between the proper connector terminals as specified on the schematic diagram.

3.5.2 DIELECTRIC STRENGTH. All cable assemblies, including mating connectors, utilizing wire in accordance with MIL-W-76, shall be capable of passing the high potential test of this specification.

3.5.2.1 Cable assemblies utilizing wire in accordance with MIL-W-16878 shall be capable of satisfactory performance under the high potential test of this specification.

3.5.3 ENVIRONMENTAL REQUIREMENTS. The cable assembly shall be capable of satisfactory performance during or after, as applicable exposure to the following conditions:

a. Temperatures from -54°C (-65°F) to 74°C (165°F).

b. Shock incident to transportation, handling and service use.

c. Vibration incident to transportation and service use.

d. Salt fog likely to be encountered during storage, transportation or service use on or near a large body of salt water.

e. Humidity incident to storage or use in damp climate.

3.6 DIMENSIONING. The cable length shall be specified from cable clamp to cable clamp or cable potting. Branch length shall be specified from the clamp to the point in the junction where the branch cables are effectively prevented from separating. General cable dimensioning and tolerances shall be in accordance with figure 4.

3.7 PRODUCT MARKINGS

3.7.1 LABELING. The main cable, branch cables, connectors, legs, and other terminations shall be positively identified by means of clearly marked tape labels secured to the cable beneath the external sleeve, or by means of hot stamp markings with Kingsely Hot Stamp Machines or equal, on the insulation or external sleeve. Each label shall carry no less than 2 identification markings parallel to the cable. The information shown on the label shall be in accordance with MIL-STD-130.

3.7.1.1 LABEL MATERIAL. When tape labels are used, they shall be made of polyvinyl chloride.

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3.7.1.2 LOCATION OF LABELS. Labels shall be located in compliance with figure 4.

3.7.1.3 STAMPING SIZE. The stamped marking shall be of the largest size commensurate with the cable or wire diameter, but in no case shall it exceed 1/4 inch.

3.7.1.4 STAMPING COLOR. The color of the marking shall contrast with that of the tape of sleeve. Yellow is the preferred color for opaque tape; black is preferred for all other marking.

3.7.1.5 STAMPING QUALITY. The marking shall be legible, permanent, and durable. Cover-coating of hot-stamp marking is permitted.

3.7.2 NOMENCLATURE. Equipment nomenclature shall be assigned on the basis of approved name and item descriptions. Approved nomenclature for each major component will be furnished upon application by the contractor to the procuring activity, at such time as all identifying details are available to the contractor in each production item, with a description of each item in accordance with MIL-D-26715. The approved nomenclature will be used in connection with nameplates, handbooks, shipping records, and other official documents as may be required in the contract.

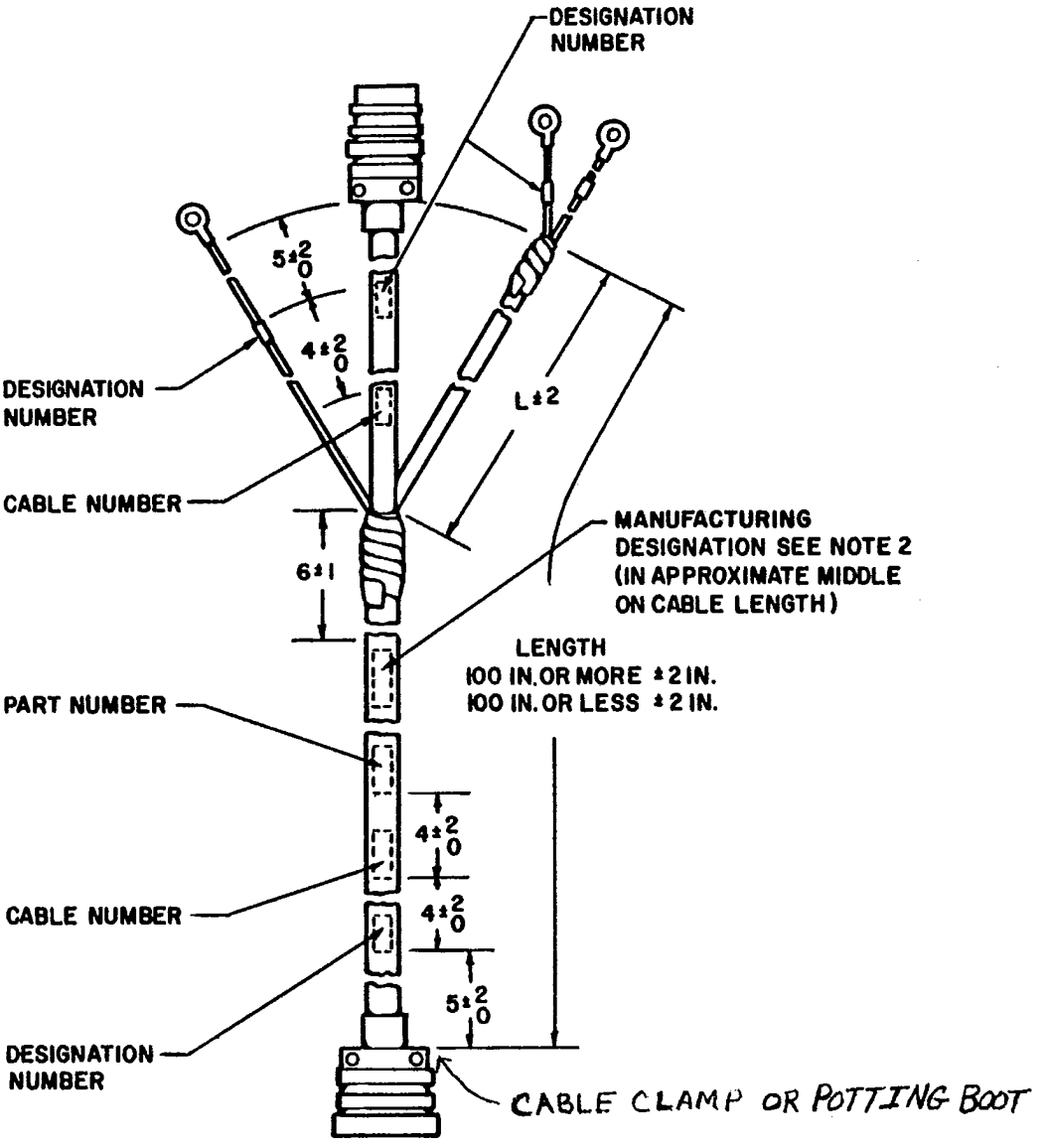
3.7.2.1 APPROVED NOMENCLATURE. The appearance of nomenclature in invitations to bids, contracts, specifications and drawings, does not necessarily constitute approved nomenclature. Prior to the use of nomenclature, the contractor is required to obtain approved nomenclature or confirmation of the nomenclature from the procuring activity. Assignment of nomenclature on previous contracts does not in itself relieve the contractor of the requirements herein.

3.8 WORKMANSHIP. The cable assembly shall be assembled and finished in a careful manner so that no degradation of the materials shall be caused by poor workmanship. Particular attention shall be given the individual wires to insure the prevention of kinks and insulation damage during assembly.

3.8.1 WIRE STRIPPING. The insulation of each wire shall be stripped back no farther than is necessary for a secure joint, and insofar as practicable, no individual strand shall be damaged by the stripping process. Allowable strand damage shall not exceed the following rate:

NUMBER OF STRANDS IN CONDUCTOR	MAXIMUM NUMBER OF DAMAGED STRANDS
0-6	o
7-16	1
17-26	2
27-36	3
37-46	4
More than 46	10% total strands

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NOTE: 1. DIMENSIONS IN INCHES
2. MANUFACTURING INFORMATION SHALL INCLUDE MANUFACTURERS NAME OR CODE AND THE MONTH AND YEAR OF ASSEMBLY

FIGURE 4 GENERAL CABLE DIMENSIONING AND LABELING

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3.8.2 SOLDERING. The soldering process ahll be in accordance with MIL-S-6872. Solder joints shall provide a positive electrical and a satisfactory mechanical connection.

3.8.3 LACING. The lacing cord shall be pulled snug but not excessively tight, and the ends shall be secured without knots. The lacing length shall be a minimum of 10 close-spaced turns, but not less than 3/8 inch. Care shall be exercised that lacing does not cause tension on the connector pins.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. The inspection and testing of the cable assemblies shall be classified as follows:

Acceptance Tests

4.2 ACCEPTANCE TESTS. Acceptance tests shall consist of:

a. Individual Tests

b. Sampling Plans and Tests

4.2.1 INDIVIDUAL TESTS. Each cable assembly shall be subjected to the following tests as described under "4.4 TEST METHODS" of this specification:

a. Examination of Product

b. Continuity Test

c. High Potential Test

4.2.2 SAMPLING PLANS AND TESTS

4.2.2.1.1 SAMPLING IN ACCORDANCE WITH MIL-STD-105. Unless otherwise specified in the contract or purchase order for the cable assemblies to be delivered, any sampling procedure used shall be in accordance with MIL-STD-105. AQL and Inspection Level shall be as required by the detail specification or drawings.

4.2.2.1.2 REJECTION AND RETEST. When the sample item from a lot fails to meet the specified tests, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

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4.2.2.1.2.1 INDIVIDUAL TESTS MAY CONTINUE. For operational reasons, individual tests may be continued pending the the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.2.2.2 SAMPLING TESTS. The following special environmental tests shall be conducted on finished cable assemblies which have been set aside for sampling test by the sampling process:

a. Heat Resistance Test

b. Cold Test

c. Shock Test

d. Vibration Resistance Test

e. Humidity Test

4.2.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a sampling test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS. The test conditions are described under the individual tests to which they apply.

4.4. TEST METHODS

4.4.1 EXAMINATION OF PRODUCT. Each finished cable assembly shall be inspected to determine compliance with the workmanship requirements of this specification.

4.4.2 CONTINUITY TEST. A milli-ohmmeter capable of measuring the resistances derived from Table 1 shall be used to perform this test. Measuring tolerance for this instrument shall be plus or minus 3 percent. The resistance of each circuit shall be measured. A measured resistance more than 10 percent greater than the maximum resistance derived from Table 1 shall be cause for rejection. Maximum resistance shall be determined as follows:

MR = N(CD) + LR
N = Number of conductors in series
CD = Contact resistance in milli-ohms
L = Length of conductor in feet
R = Resistance of conductor in milli-ohms per foot

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TABLE I

MAXIMUM RESISTANCE PER CIRCUIT AT 25°C (77°F)

Conductor Size	Connector Contact Size	Resistance Of Connector (Milli-Ohms)	Resistance Of Conductor (Milli-Ohms per ft)
20	16	1	10.5
18	16	1	6.6
16	16	1	5.0
16	12	1	3.0

4.4.3 HIGH POTENTIAL TEST

4.4.3.1 CABLES UTILIZING WIRE IN ACCORDANCE WITH MIL-W-76. A 60-cps test voltage of magnitude specified in Table II shall be successively applied to each wire in the following manner:

4.4.3.1.1 POINTS OF APPLICATION. When applicable, test voltages form Table II shall be applied to one wire through the mating connector or termination with the remaining wires grounded to the return circuit.

4.4.3.1.2 DURATION OF APPLICATION. The duration of application shall be not less than 5 seconds from the time the specified maximum voltage is attained.

4.4.3.1.3 Each cable shall withstand the applicable test voltage specified in Table II for the duration of time specified in 4.4.3.1.2. Failure to withstand the test voltage for the specified time shall be cause for rejection.

TABLE II

DIELECTRIC STRENGTH: MIL-W-76 WIRE

CONNECTOR SERVICE CLASS	RMS TEST VOLTAGE FOR WIRE TYPE MW	HW
INST.	950	950
A	1900	1900
D	2800	2800
E	3000	3500
B	3000	4500
C	3000	6000

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4.4.3.2 CABLES UTILIZING WIRE IN ACCORDANCE WITH MIL-W-16878. Cables utilizing wire in accordance with MIL-W-16878 shall be examined in accordance with procedures prescribed therein.

4.4.4 ENVIRONMENTAL TESTS. During or after, as applicable, the representative cable assembly has been subjected to an environmental test it shall meet the requirements of the specified check test. Failure to meet these requirements shall be considered as failure to pass the environmental test and cause for rejection.

4.4.4.1 HEAT RESISTANCE TEST. The finished cable assembly shall be placed in a suitable chamber and maintained at 74°C (165°F) $\pm$ 5°F at less than 15 percent relative humidity for a period of 72 hours. When cooled to room temperature the cables shall be subjected to the high potential test.

4.4.4.2 COLD TEST. The finished cable assembly shall be placed in a suitable chamber and maintained at a temperature of -54 (-65°F) $\pm$ 5°F for a period of 72 hours. At the end of this period the cable assembly shall be allowed to return to room temperature and any existing moisture removed from the connectors. The cable assembly shall then be subjected to the high potential test.

4.4.4.3 SHOCK TEST

4.4.4.3.1 PREPARATION. The finished cable assembly shall be rack mounted to simulate service conditions with all lugs, connectors and other terminations secured to mating parts. Provisions shall be made for securing the central portions of the cable to the rack to prevent travel or damage during the test. Long cable lengths may be coiled.

4.4.4.3.2 PROCEDURE. The cable shall be subjected to a shock test in accordance with Method 516, procedure 1 of MIL-STD-810. At conclusion of the shock test the continuity test shall be performed.

4.4.4.4 VIBRATION RESISTANCE TEST. The finished cable assembly shall be examined for vibration effects as follows:

4.4.4.4.1 PREPARATION. The cable assembly shall be rack mounted to simulate use conditions with all lugs, connectors, and other terminations secured to mating parts. Provisions shall be made for securing the central portions of the cable to the rack to prevent travel during the test. Long cable lengths may be coiled. The cable assembly, on the mounting rack, shall be subjected to a vibration test in accordance with Method 514, procedure 1 of MIL-STD-810. Following the vibration period the specimen shall be subjected to the continuity test.

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4.4.4.5 SALT FOG TEST. The finished cable assembly connected to mating connectors to simulate service conditions, shall be subjected to a salt fog test in accordance with method 509, Procedure 1, MIL-STD-810. At the conclusion of the salt fog test, the high potential test shall be performed.

4.4.4.6 HUMIDITY TEST. The resistance to moisture of the finished cable assembly shall be determined by subjecting the assembly to five humidity cycles.

4.4.4.6.1 PREPARATION. The cable assembly shall be dried in a 55°C 130°F oven at less than 15 percent relative humidity for 24 hours and then mounted in the test chamber on a rack simulating use conditions. This rack shall be designed to facilitate performance of the high potential test without removal from the test chamber. Following the initial high potential test, the cable assembly shall be subjected to five of the following 48-hour humidity cycles.

4.4.4.6.2 HUMIDITY CYCLE. Beginning with zero hour the test chamber temperature cycles shall conform to the schedule in Table III with the relative humidity maintained at 90 to 98 percent.'

TABLE III

HUMIDITY TEST CYCLE

TEST CYCLE AT 90 to 98 PERCENT RELATIVE HUMIDITY

STEP	PERIOD HOURS	TEMPERATURE EXCURSION °F	°F ± 5° CONSTANT
1	0-4	88-149	149
2	4-12		
3	12-16	149-88	88
4	16-24 24-28 28-37	high potential test	88
5	37-38	88-68	
6	38-42		
7	42-43	68-88	
8	43-48	high potential test	88

4.4.4.6.3 MEASUREMENTS. The high potential test shall be performed during step 4 and 8 of each cycle.

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5. PREPARATION FOR DELIVERY

5.1 PACKAGING. Packaging shall be such as to insure delivery of the finished product in operable condition by any common carrier and as specified in each detail specification, contract or purchase order to the Government.

5.2 SHIPPING. Shipping instructions and point of delivery will be furnished by the procuring activity.

5.3 MARKING. Marking of the shipping containers shall be in accordance with MIL-STD-129. The shipment marking nomenclature shall be as specified in the detail specification.

6. NOTES

6.1 INTENDED USE. This specification is intended for use to incorporate, by reference, in detail equipment specifications or drawings, those requirements which are common to the design, manufacture, and test requirements for cable assemblies which are part of, or used with, special weapons and associated airborne and ground test equipment. This specification does not cover RF cables.

6.2 ORDERING DATA. Ordering data for one type of special weapons cable assembly shall be as specified in the detail specification of drawings.

Custodian
Air Force - 15

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Air Force - 79

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Air Force - 15

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